The tree of rationals generated by $x \mapsto x+1$ and $x \mapsto -1/x$: an exact distance formula and the rank recurrence

A solution to Problem 137 of Kagey's Open Problem Collection

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Abstract

Start at 0 and keep applying the maps $x \mapsto x+1$ and $x \mapsto -1/x$, the second one only to nonzero numbers. Every rational is reached, and its rank is the least number of steps needed. Sorting the rationals by rank gives the tree in OEIS A226247/A226248. Kagey asks whether the number $a(n)$ of rationals of rank n satisfies $a(n) = a(n-1) + a(n-3)$ for $n \geq 4$, whether the vertices reached last by $x \mapsto -1/x$ are exactly the negative ones, and whether the rationals of a given rank can be described directly. We answer all 3 questions, and each answer is yes. The main tool is an exact formula for the rank: a positive rational with negative (Hirzebruch–Jung) continued fraction digits $c_1, \ldots, c_k$ has rank $c_1 + \cdots + c_k + k - 1$, and a negative x has rank one more than $-1/x$. From this we get the tree structure, a description of each rank by compositions, and the generating function $(1 + t^2 - t^3)/(1 - t - t^3)$ of $a(n)$. This also fills the gap in the accepted Math.SE answer, which counted words in the 2 maps but did not show that these words match the rationals one to one.

1 Introduction

1.1 Kagey's questions

Suppose we start at 0 and may make 2 kinds of moves: add 1, or replace a nonzero number x by $-1/x$. We write

$$f(x) = x + 1, \qquad g(x) = -\frac{1}{x}.$$

From 0 the only move is f, to 1. From 1 we can move to 2 or to -1, and so on. Every number we reach is rational, and every rational is reached (Theorem 3.6). This is not a surprise, since f and g generate the modular group $\mathrm{PSL}_2(\mathbb{Z}) \cong C_2 * C_3$, which acts on the rational projective line [12, 13]. The *rank* of a rational x is the least number of moves that takes 0 to x. Kimberling [4] sorted the rationals by rank in 2013. The result is the tree of OEIS A226247 (denominators) and A226248 (numerators) [4, 5], drawn in Figure 1. Let $a(n)$ be the number of rationals of rank n. The sequence starts $1, 1, 2, 2, 3, 5, 7, 10, 15, 22, 32, 47, 69, 101, \ldots$, and Kimberling's entry says, without a proof, that it is A097333 shifted by one [7].

Kagey asked about this sequence on Mathematics Stack Exchange in April 2025 [2], and it is Problem 137 of his Open Problem Collection [1]. He colors a rational red if the last move on the way to it is f, and blue if it is g, and asks 3 questions.

Q1. Is $a(n) = a(n-1) + a(n-3)$ for all $n \geq 4$?

Q2. Is a vertex blue if and only if its value is negative?

Q3. Is there a way to characterize the rationals of rank n?

We answer all 3 questions, and each answer is yes. The recurrence holds for every $n \geq 4$, and it fails at $n = 3$ (Theorem A). A shortest way to reach a positive x always ends with the move from $x - 1$, and one to reach a negative x always ends with the move from $-1/x$, so blue means negative (Theorem C). The rank of a rational can be read off its continued fraction, so each rank can be listed directly (Theorem B).

1.2 The first ranks by hand

Before any theory, here are ranks 0 to 5, worked out by hand. Rank 0 is just 0. To get rank n, we apply f and g to every rational of rank $n - 1$ and keep the values we have not seen before. Table 1 shows the work, and Figure 1 shows ranks 0 to 7 as a tree.

n	moves from the rationals of rank $n-1$	rank n	$a(n)$
0	(start)	0	1
1	$0 \mapsto 1$ ($g(0)$ is not defined)	1	1
2	$1 \mapsto 2,\ -1$	$2,\ -1$	2
3	$2 \mapsto 3,\ -\frac{1}{2}$; $-1 \mapsto 0_{[0]},\ 1_{[1]}$	$3,\ -\frac{1}{2}$	2
4	$3 \mapsto 4,\ -\frac{1}{3}$; $-\frac{1}{2} \mapsto \frac{1}{2},\ 2_{[2]}$	$4,\ -\frac{1}{3},\ \frac{1}{2}$	3
5	$4 \mapsto 5,\ -\frac{1}{4}$; $-\frac{1}{3} \mapsto \frac{2}{3},\ 3_{[3]}$; $\frac{1}{2} \mapsto \frac{3}{2},\ -2$	$5,\ -\frac{1}{4},\ \frac{2}{3},\ \frac{3}{2},\ -2$	5

Table 1: Ranks 0 to 5. In row n, "$x \mapsto u,\ v$" means $f(x) = u$ and $g(x) = v$ for a rational x of rank $n - 1$. Gray values were found earlier, and the subscript in brackets is their rank.

Already in these small cases we can see the answers.

(a) (Q2) Every positive rational in Table 1 is reached from $x - 1$ by f, and every negative one from $-1/x$ by g. So red means positive and blue means negative.

(b) (Q1) A positive rational has 2 new images, a rational in $(-1, 0)$ has 1, and a rational ≤ -1 has none. Since $g(g(x)) = x$, the g-image of a negative x is the rational x came from. The other old values come from applying f to some $x \leq -1$: $f(-1) = 0$ is the root, and $f(-2) = -1$ has rank 2, which is 3 ranks before -2 (Figure 1 shows 2 more such pairs). This jump back by 3 ranks is where the term $a(n - 3)$ comes from.

(c) (Q1) The counts $1, 1, 2, 2, 3, 5$ satisfy $a(4) = a(3) + a(1)$ and $a(5) = a(4) + a(2)$, but $a(3) = 2$ is one less than $a(2) + a(0)$. The defect comes from the root, the only rational where g cannot be applied (Remark 8.2).

(d) (Q3) The positive rationals of rank 5 are 5, $\frac{2}{3} = 1 - \frac{1}{3}$ and $\frac{3}{2} = 2 - \frac{1}{2}$. Expansions of this kind are the key to Q3.

The rest of the paper proves that these patterns hold at every rank. Theorem C below is the precise form of (a) and (b).

1.3 Results

We write $d(x)$ for the rank of x and X_n for the set of rationals of rank n, so that $a(n) = |X_n|$ (Definition 2.1). Rationals are always in lowest terms p/q with $q \geq 1$. For integers $c_1, \ldots, c_k$ the

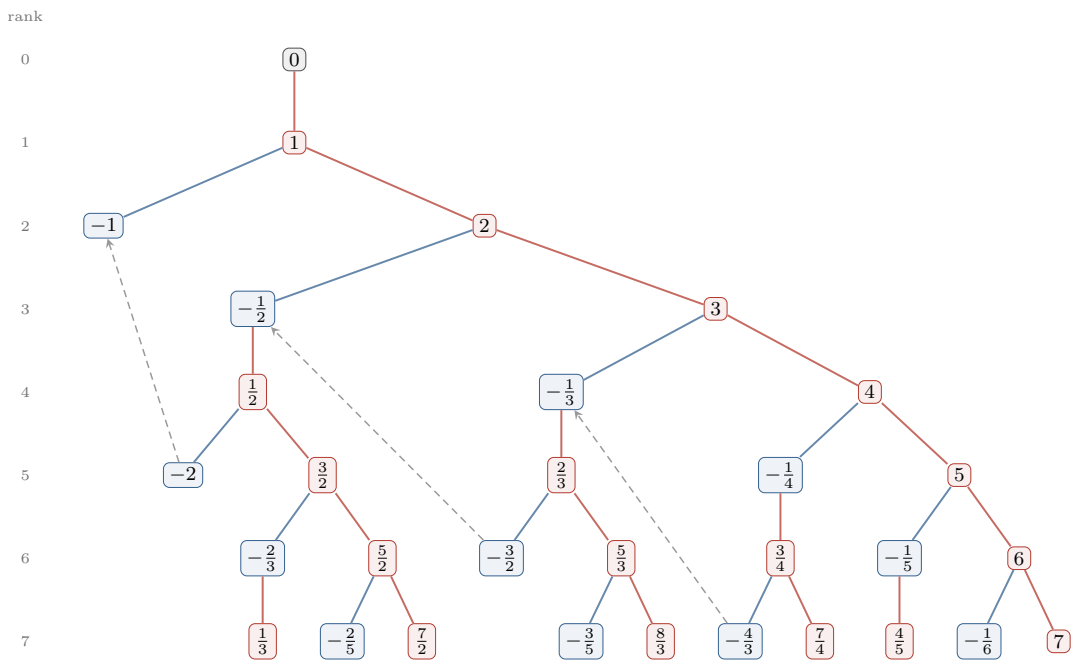


Figure 1: Ranks 0–7 of the tree. Red edges apply $f(x) = x + 1$ and blue edges apply $g(x) = -1/x$. Every positive vertex is entered by an f-edge and every negative vertex by a g-edge (Theorem C), and the vertices ≤ -1 are leaves. The dashed gray arrows show f applied to the leaves $x < -1$: the value $x + 1$ is not new, and its rank is 3 less than that of x. The numbers on the left are ranks. The rows are Kimberling's generations [4], with his generation j being rank $j - 1$.

negative continued fraction is

$$[c_1, c_2, \ldots, c_k]^- = c_1 - \cfrac{1}{c_2 - \cfrac{1}{\ddots - \cfrac{1}{c_k}}},$$

so that $[1, 3]^- = \frac{2}{3}$ and $[2, 2]^- = \frac{3}{2}$. The ordinary continued fraction $[b_0; b_1, \ldots, b_m]$ is the same with plus signs instead of minus signs, with $b_0 \ge 0$ and $b_1, \ldots, b_m \ge 1$.

Theorem A (answers Q1; = Theorem 5.1 and Proposition 5.2). The generating function of $a(n)$ is

$$\sum_{n \ge 0} a(n)\, t^n \;=\; \frac{1 + t^2 - t^3}{1 - t - t^3}.$$

Hence $a(n) = a(n-1) + a(n-3)$ for all $n \ge 4$. It is false at $n = 3$, since $a(3) - a(2) - a(0) = -1$, so we cannot lower the 4. Also $a(n) = \text{A097333}(n-1)$ for $n \ge 1$, the number of *positive* rationals of rank n is Narayana's cows sequence $\text{A000930}(n-1)$, and

$$a(n) \;=\; C\,\psi^n + O\Big(\psi^{-n/2}\Big), \qquad C = \frac{\psi^2 + \psi}{\psi^2 + 3} = 0.7019310679\ldots,$$

where $\psi = 1.4655712318\ldots$ (the "supergolden ratio") is the real root of $\psi^3 = \psi^2 + 1$.

Theorem B (the rank formula, answers Q3; = Theorems 3.6, 4.2 and 4.8). Define the parent map $P(x) = x - 1$ for $x > 0$ and $P(x) = -1/x$ for $x < 0$. For every rational x, iterating P leads to 0, and $d(x)$ is the number of steps it takes. More precisely,

(i) $d(0) = 0$, and $d(x) = d(-1/x) + 1$ for $x < 0$,

(ii) if $x > 0$ has negative continued fraction expansion $x = [c_1, \ldots, c_k]^-$ with $c_1 \geq 1$ and $c_2, \ldots, c_k \geq 2$ (it exists and is unique by Lemma 4.1), then

$$d(x) \;=\; c_1 + c_2 + \cdots + c_k \;+\; k - 1,$$

(iii) if $x > 0$ is written as the unique ordinary continued fraction $x = [b_0; b_1, \ldots, b_m]$ with m *odd* (Lemma 4.6), then

$$d(x) \;=\; \sum_{i \text{ even}} b_i \;+\; 3 \sum_{i \text{ odd}} b_i \;-\; 2.$$

In particular the rank-n rationals are listed by integer compositions (Corollary 4.3), which answers Q3.

In words: to find the rank of a positive rational, expand it as a negative continued fraction. Each digit c costs c moves, and each of the $k - 1$ places where one digit ends and the next begins costs 1 more. For example $\frac{3}{2} = [2, 2]^-$ has rank $2+2+1 = 5$. In terms of compositions, the positive rationals of rank n match the ordered sums $n = s_1 + \cdots + s_k$ with $s_1 \geq 1$ and $s_2, \ldots, s_k \geq 3$, via $s_1 = c_1$ and $s_i = c_i + 1$; for $n = 5$ the sums 5, $1 + 4$ and $2 + 3$ give 5, $\frac{2}{3}$ and $\frac{3}{2}$.

Theorem C (the tree, answers Q2; = Corollaries 3.7 and 3.8). The only in-neighbour of $x \neq 0$ that is closer to the root is $P(x)$. So Kimberling's construction really is a tree, and no value shows up twice, not even twice in the same generation. The edge into x is an f-edge when $x > 0$ and a g-edge when $x < 0$, so *a vertex is blue if and only if its value is negative*, which answers Q2. A vertex $x > 0$ has 2 children, $x + 1$ and $-1/x$. A vertex $-1 < x < 0$ has 1 child, $x + 1$. A vertex $x \leq -1$ is a leaf, and for $x < -1$ the value $x + 1$ is not new but appeared exactly *3 ranks earlier*.

1.4 How the proof goes, and what was missing before

The parent map P of Theorem B undoes the last move guessed in observation (a). Section 3 shows that P traces a shortest path back to 0 and answers Q2, Section 4 counts its steps with continued fractions and answers Q3, and Section 5 counts the ranks and answers Q1.

The accepted answer to [2], by the user mathmasterzach [3], got most of the way on Q1: it counted the words in f and g that avoid certain patterns, and got the right generating function. But that counts *words*, not *rationals*, and the answer does not show that the 2 match one to one. Section 6 supplies this step. Section 2 fixes the notation, Section 7 lists the computer checks and Section 8 has a few remarks. None of the proofs relies on the references.

2 Setting and notation

Rationals are written in lowest terms p/q with $q \geq 1$, and q is the *denominator*. For a real x, $\lceil x \rceil$ is the least integer $\geq x$ and $\lfloor x \rfloor$ the greatest integer $\leq x$. We write $\mathbb{Q}_{>0}$ and $\mathbb{Q}_{<0}$ for the positive and negative rationals and $|S|$ for the number of elements of a finite set S.

Definition 2.1. Let G be the directed graph on $\mathbb{Q}$ with an *f-edge* $x \to x+1$ for every x, and a *g-edge* $x \to -1/x$ for every $x \neq 0$. There is no g-edge out of 0, since $g(0)$ is not defined. If $y \to x$ is an edge, y is an *in-neighbour* of x and x an *out-neighbour* of y. For $x \in \mathbb{Q}$ let

$$d(x) \;=\; \text{length of a shortest directed path from 0 to } x \text{ in } G \;\in\; \mathbb{Z}_{\geq 0} \cup \{\infty\},$$

and call $d(x)$ the *rank* of x. In other words, $d(x)$ is the smallest number of times we have to apply f and g to get from 0 to x, never applying g to 0. As in [2], we write

$$X_n = \{x \in \mathbb{Q} \mid d(x) = n\}, \qquad a(n) = |X_n|, \qquad r(n) = |X_n \cap \mathbb{Q}_{>0}|.$$

Theorem 3.6 will show that $d(x) < \infty$ for all x, and Corollary 4.3 that every X_n is finite.

Note that $d(0) = 0$ and $X_0 = \{0\}$, since the only path of length 0 is the empty one.

Words. We write a path as a word in f and g, read left to right in the order the moves are made, so fg takes 0 to 1 and then to -1. (As a composition, $\ell_1\ell_2\cdots\ell_j$ is $\ell_j \circ \cdots \circ \ell_1$.) A word is *admissible at* 0 if it never applies g to 0. These words are the paths in G from 0, so $d(x)$ is the length of a shortest admissible word that takes 0 to x.

Kimberling's generations. Kimberling [4] builds rows as follows. Row 1 is (0). If row n is $(c_1, \ldots, c_z)$, write down $(c_1+1,\ -1/c_1,\ c_2+1,\ -1/c_2,\ \ldots)$, leaving out $-1/c_i$ when $c_i = 0$, and delete anything that is already in rows $1, \ldots, n$. What is left is row $n+1$. Each value in row $n+1$ gets the color of the map that produced it. This is breadth-first search from 0, so the rows should be our sets X_n. The next proposition checks this. Kagey's ranks [1] are the same as well.

Proposition 2.2 (generations are metric spheres)**.** *For every $n \geq 0$, the set of values in Kimberling's row $n+1$ equals X_n.*

Proof. We use induction on n. For $n = 0$ both sides are $\{0\}$. Let $n \geq 1$ and suppose that, for every $j < n$, the values in row $j+1$ form the set X_j. Take y in row $n+1$. Then y is $f(x)$ or $g(x)$ for some x in row n, that is, for some $x \in X_{n-1}$, so $d(y) \leq n$. Since y was not deleted, it is not in $X_0 \cup \cdots \cup X_{n-1}$, so $d(y) \geq n$. Hence $y \in X_n$. Conversely, let $y \in X_n$ and let $x \to y$ be the last edge of a shortest path from 0 to y. The rest of the path is a path of length $n-1$ from 0 to x, so $d(x) \leq n-1$, and $d(x) \geq n-1$ because $d(y) \leq d(x)+1$. So $x \in X_{n-1}$ is in row n, and y is written down when we form row $n+1$. It is not deleted, because $d(y) = n$ means that y is not in $X_0 \cup \cdots \cup X_{n-1}$. □

Remark 2.3. A value y could in principle be written down *twice* when forming row $n+1$, say as $c+1$ and as $-1/c'$. The construction does not say what to do with the second copy, and the color of y would not be well defined. Corollary 3.8(a) shows that this never happens, so the rows have no repeats, $|\text{row } n+1| = a(n)$, and the colors are well defined.

3 The parent map and the exact distance formula (Q2)

Here is the idea of this section. Observation (a) of Section 1.2 suggests a guess for a shortest path to x: undo the last move, from $x-1$ if $x > 0$ and from $-1/x$ if $x < 0$, and repeat. Let $D(x)$ be the number of undo steps needed to get back to 0. The guessed path has length $D(x)$, so $d(x) \leq D(x)$. For the reverse inequality it is enough to check that D *goes up by at most* 1 *along every edge of* G:

then a path of length n from 0, where $D = 0$, ends at a point with $D \le n$. Along the guessed edges D goes up by exactly 1, by definition. A g-edge into a positive number is easy, because g undoes itself. The heart of the matter is an f-edge into a negative x. It comes from $x - 1$, and we will see that $D(x-1) = D(x) + 3$. For example $D(-\frac{1}{2}) = 3$, from $-\frac{1}{2} \to 2 \to 1 \to 0$, and $D(-\frac{3}{2}) = 6$, from $-\frac{3}{2} \to \frac{2}{3} \to -\frac{1}{3} \to 3 \to 2 \to 1 \to 0$. This is the jump back by 3 ranks from observation (b).

Definition 3.1. The *parent map* $P : \mathbb{Q} \setminus \{0\} \to \mathbb{Q}$ is

$$P(x) = \begin{cases} x - 1, & x > 0, \\ -1/x, & x < 0. \end{cases}$$

Note that P undoes an edge of G. For every $x \neq 0$ there is an edge $P(x) \to x$, which is the f-edge $(x-1) \to x$ when $x > 0$ and the g-edge $(-1/x) \to x$ when $x < 0$. The second one is allowed because $-1/x \neq 0$.

Lemma 3.2 (termination)**.** *For every $x \in \mathbb{Q}$ the sequence x, $P(x)$, $P^2(x), \ldots$ reaches 0 after finitely many steps.*

Proof. First let $x > 0$. The values $x, x-1, \ldots, x - \lceil x \rceil + 1$ are positive, so P subtracts 1 a total of $\lceil x \rceil$ times and reaches $x - \lceil x \rceil$. This is 0 if $x \in \mathbb{Z}$ and lies in $(-1, 0)$ otherwise, and it has the same denominator as x. Next, if $z \in (-1, 0)$, write $z = -r/q$ in lowest terms with $0 < r < q$. Then $P(z) = -1/z = q/r$ is bigger than 1 and has the smaller denominator r.

Now we use induction on the denominator q to prove the lemma for $x \in \mathbb{Q}_{>0} \cup (-1, 0)$. If $q = 1$, then x is a positive integer (no element of $(-1, 0)$ has denominator 1), and it reaches 0 after x steps. Let $q \ge 2$. If $x > 0$, then x is not an integer, so its sequence reaches a point of $(-1, 0)$ with denominator q. The next point is positive with denominator less than q, so we are done by induction. If $x \in (-1, 0)$, then $P(x)$ is positive with denominator less than q, and again we use induction.

Finally, if $x \le -1$ then $P(x) = -1/x > 0$, which is the positive case, and $x = 0$ is clear. □

Definition 3.3. For $x \in \mathbb{Q}$ let $D(x)$ be the number of steps after which the sequence x, $P(x)$, $P^2(x)$, … first reaches 0. Thus $D(0) = 0$ and

$$D(x) = D(P(x)) + 1 \qquad (x \neq 0), \tag{1}$$

and these two properties determine D, as the next lemma shows.

Lemma 3.4 (uniqueness principle)**.** *If $F : \mathbb{Q} \to \mathbb{Z}$ satisfies $F(0) = 0$ and $F(x) = F(P(x)) + 1$ for all $x \neq 0$, then $F = D$.*

Proof. We show $F(x) = D(x)$ by induction on $D(x)$, which is finite by Lemma 3.2. If $D(x) = 0$ then $x = 0$ and $F(0) = 0$. If $D(x) \ge 1$ then $x \neq 0$ and $D(P(x)) = D(x) - 1$, so by induction $F(x) = F(P(x)) + 1 = D(P(x)) + 1 = D(x)$. □

Next we see how D changes along each of the 4 kinds of edges of G: f-edges and g-edges, each into a positive or a negative number (the only edge into 0, from -1, is trivial). (D1) and (D2) are the 2 kinds that P undoes, and (D3) and (D4) are the other 2. The first 3 are easy, and (D4) is where the work is.

Lemma 3.5 (increment identities)**.** *For all $x \in \mathbb{Q}$ the following hold.*

(D1) If $x > 0$, then $D(x) = D(x-1) + 1$.

(D2) If $x < 0$, then $D(x) = D(-1/x) + 1$.

(D3) If $x > 0$, then $D(-1/x) = D(x) + 1$.

(D4) If $x < 0$, then $D(x-1) = D(x) + 3$.

Proof. (D1) and (D2) are just (1). For (D3), $-1/x < 0$, so (D2) applied to $-1/x$ gives $D(-1/x) = D\Big(-\frac{1}{-1/x}\Big) + 1 = D(x) + 1$.

For (D4), put $z = x - 1$. Since $x < 0$, we have $z < -1$. We follow the sequence of z for 4 steps, checking the sign each time to see which case of P applies. The sequence is

$$z \overset{P}{\longmapsto} \frac{1}{1-x} \overset{P}{\longmapsto} \frac{x}{1-x} \overset{P}{\longmapsto} 1 - \frac{1}{x} \overset{P}{\longmapsto} -\frac{1}{x} = P(x).$$

Indeed, $z < 0$, so $P(z) = -1/z = 1/(1-x)$, which is in $(0,1)$ because $1 - x > 1$. Then $P^2(z) = \frac{1}{1-x} - 1 = \frac{x}{1-x}$, which is negative because $x < 0 < 1 - x$. Then $P^3(z) = -\frac{1-x}{x} = 1 - \frac{1}{x}$, which is bigger than 1 because $-1/x > 0$. Finally $P^4(z) = -\frac{1}{x}$, and this is $P(x)$ because $x < 0$. None of these 4 points is 0, so $D(z) = 4 + D(P(x)) = 4 + \big(D(x) - 1\big) = D(x) + 3$. □

Theorem 3.6 (exact distance formula, dynamic form)**.** *Every rational is reachable from 0 in G, and $d(x) = D(x)$ for all $x \in \mathbb{Q}$.*

In words: the path traced by the parent map is a shortest path. So the rank of x is the number of parent steps from x back to 0, and this number is easy to compute. The idea of the proof was given at the start of this section.

Proof. Step 1: the parent path, so $d \le D$. Let $x \in \mathbb{Q}$. The reversed sequence $0 = P^{D(x)}(x)$, $P^{D(x)-1}(x)$, …, $P(x)$, x is a directed path in G, because each consecutive pair $(P(y), y)$ is an edge, as noted after Definition 3.1. It has length $D(x)$, so x is reachable and $d(x) \le D(x)$.

Step 2: D goes up by at most 1 along each edge. We check that

$$D(x) \le D(y) + 1 \qquad \text{for every edge } y \to x \text{ of } G. \tag{2}$$

There are 5 cases.

- For an f-edge into $x > 0$, $y = x - 1$ and (D1) gives $D(x) = D(y) + 1$.
- For an f-edge into $x = 0$, $y = -1$ and $D(0) = 0 \le D(-1) + 1$.
- For an f-edge into $x < 0$, $y = x - 1$ and (D4) gives $D(x) = D(y) - 3 \le D(y) + 1$.
- For a g-edge into $x < 0$, $y = -1/x$ and (D2) gives $D(x) = D(y) + 1$.
- For a g-edge into $x > 0$, $y = -1/x < 0$ and (D3) gives $D(y) = D(x) + 1$, so $D(x) = D(y) - 1$.

There is no g-edge into 0, since $-1/y \neq 0$.

Step 3: $d \ge D$. Let $x \in \mathbb{Q}$. By Step 1, $n = d(x)$ is finite, so there is a path $0 = y_0 \to y_1 \to \cdots \to y_n = x$. By (2), $D(y_i) \le D(y_{i-1}) + 1$ for $i = 1, \ldots, n$, and so $D(x) \le D(0) + n = d(x)$. □

From now on we use d and D interchangeably. The next 2 corollaries describe the tree near each vertex, and together they prove Theorem C.

Corollary 3.7 (in-neighbour distances)**.** *Let $x \neq 0$. The in-neighbours of x in G are $x - 1$ (by f) and $-1/x$ (by g), with ranks as in the table.*

	in-neighbour $x-1$	in-neighbour $-1/x$
$x>0$	$d(x)-1$	$d(x)+1$
$x<0$	$d(x)+3$	$d(x)-1$

In particular $P(x)$ is the only in-neighbour of x that is closer to the root than x.

Proof. These are the in-neighbours because $f(y)=x$ only for $y=x-1$, and $g(y)=x$ only for $y=-1/x$, which is nonzero. The table is (D1)–(D4) together with Theorem 3.6. Since the 2 in-neighbours have different ranks, they are different. □

Corollary 3.8 (tree structure, answering Q2)**.** *(a) For every $x\neq 0$, the parent $P(x)$ is the only rational of rank $d(x)-1$ from which a single move reaches x. In Kimberling's construction each $x\neq 0$ is written down exactly once, by $P(x)$, when row $d(x)+1$ is formed. So no value is produced twice, not even twice in one generation. The construction gives, without any choices, the tree on $\mathbb{Q}$ in which the parent of each $x\neq 0$ is $P(x)$.*

(b) Every shortest path from 0 to $x\neq 0$ ends with an f-edge if $x>0$ and with a g-edge if $x<0$. So the colors of Q2 are well defined, and a vertex is blue if and only if its value is negative. *The root has no color.*

(c) The children *of x, that is, the y with $P(y)=x$, are as follows. A vertex $x>0$ has the 2 children $x+1$ and $-1/x$. A vertex $x\in(-1,0)$ has the 1 child $x+1\in(0,1)$, and the root has the 1 child* 1. *Every vertex $x\leq -1$ is a leaf. Every child of x has rank $d(x)+1$. Moreover, for $x<-1$ the value $x+1$ is not new but* 3 ranks old*, with $d(x+1)=d(x)-3$, and for $x=-1$ the value $x+1=0$ is the root.*

Proof. (a) The first sentence is Corollary 3.7. Now we show by induction on n that row $n+1$ lists each element of X_n exactly once. For $n=0$ this is clear. Let $n\geq 1$, suppose that row n lists each element of X_{n-1} once, and let $x\in X_n$. When row $n+1$ is formed, x is written down once for each entry c of row n and each move taking c to x. Such a c is an in-neighbour of x of rank $n-1$, so $c=P(x)$, which appears once in row n. And c reaches x by only one move, since $c+1=-1/c$ would mean $c^2+c+1=0$. So x is written down exactly once and kept. The other values written down are in earlier rows (Proposition 2.2) and are deleted.

(b) If $y\to x$ is the last edge of a shortest path, then $d(y)=d(x)-1$, so $y=P(x)$ by (a). The edge $P(x)\to x$ is an f-edge when $x>0$ and a g-edge when $x<0$.

(c) We solve $P(y)=x$. If $y>0$, it says $y=x+1$, which is possible exactly when $x+1>0$. If $y<0$, it says $-1/y=x$, which is possible exactly when $x>0$, with $y=-1/x$. This gives the list of children. Each child y has $d(y)=d(P(y))+1=d(x)+1$ by (1). If $x<-1$, then $x+1<0$, and (D4) applied to $x+1$ gives $d(x)=d(x+1)+3$. □

Remark 3.9. Read backwards, the computation in (D4) gives the path

$$-\tfrac{1}{x}\xrightarrow{f}1-\tfrac{1}{x}\xrightarrow{g}\tfrac{x}{1-x}\xrightarrow{f}\tfrac{1}{1-x}\xrightarrow{g}x-1,$$

so $x-1=(g\circ f\circ g\circ f\circ g)(x)$. This is the relation $f^{-1}=g\circ f\circ g\circ f\circ g$ from [1], or equivalently $(g\circ f)^3=\mathrm{id}$, which together with $g^2=\mathrm{id}$ presents $\mathrm{PSL}_2(\mathbb{Z})\cong C_2*C_3$ [12, 13]. The 3 in (D4) is the 5 letters of the relator minus the 2 that cancel in $g\cdot g=\mathrm{id}$: the shortest path to $x<0$ ends with g, and the path to $x-1$ replaces that g by $gfgfg$. This 3 gives the t^3 in Theorem A. The proof of (D4) needs no group theory.

Example 3.10. Take $x = -\frac{7}{2}$. The parent sequence is

$$-\tfrac{7}{2} \to \tfrac{2}{7} \to -\tfrac{5}{7} \to \tfrac{7}{5} \to \tfrac{2}{5} \to -\tfrac{3}{5} \to \tfrac{5}{3} \to \tfrac{2}{3} \to -\tfrac{1}{3} \to 3 \to 2 \to 1 \to 0,$$

so $d(-7/2) = 12$. Reversing it gives the shortest word $fffgffgffgfg$, which means we apply $f, f, f, g, f, f, g, f, f, g, f, g$ to 0 in this order.

4 The rank from continued fractions (Q3)

Theorem 3.6 reduces the rank to counting parent steps. Here is why continued fractions count them. The parent sequence of $\frac{2}{7}$ in Example 3.10 makes 1 subtraction, a flip, 2 subtractions, a flip, 2 subtractions, a flip, and 3 subtractions, so $d(\frac{2}{7}) = (1 + 2 + 2 + 3) + 3 = 11$. In general, from a positive x the map P subtracts 1 exactly $c = \lceil x \rceil$ times, and if x is not an integer one flip then gives $1/(c - x) > 1$. The numbers c we meet are the digits of the negative continued fraction (here $\frac{2}{7} = [1, 2, 2, 3]^-$), so the rank is the sum of the digits plus the number of flips. The negative continued fraction is also called the Hirzebruch–Jung or "minus" continued fraction [9, 10]. We prove what we need from scratch.

For integers $c_1, \ldots, c_k$ the negative continued fraction is defined by

$$[c_1]^- = c_1, \qquad [c_1, c_2, \ldots, c_k]^- = c_1 - \frac{1}{[c_2, \ldots, c_k]^-},$$

whenever the denominators are nonzero. We call $(c_1, \ldots, c_k)$ a *digit string*.

Lemma 4.1 (existence and uniqueness of the expansion)**.** *Every rational $x > 0$ has exactly one representation*

$$x = [c_1, c_2, \ldots, c_k]^-, \qquad k \geq 1, \quad c_1 \geq 1, \quad c_i \geq 2 \ (2 \leq i \leq k),$$

and in it $c_1 = \lceil x \rceil$. Moreover every tail satisfies $[c_j, \ldots, c_k]^- > 1$ for $j \geq 2$.

Proof. Existence. Given $x > 0$, write down $c = \lceil x \rceil$. If $x = c$ we stop, and otherwise we repeat with $x' = 1/(c - x)$. Suppose $x = p/q$ in lowest terms is not an integer. Then $(c-1)q < p < cq$, so $cq - p$ is between 1 and $q - 1$, and $x' = q/(cq - p)$. This is again in lowest terms, since a common divisor of q and $cq - p$ divides p, and its denominator $cq - p$ is less than q. So the process stops. Also $x' > 1$, so every digit after the first is $\lceil x' \rceil \geq 2$, and $c_1 = \lceil x \rceil \geq 1$ because $x > 0$. Using $x = c - 1/x'$ at each step shows that $x = [c_1, \ldots, c_k]^-$.

Tails are bigger than 1. We use downward induction. First $[c_k]^- = c_k \geq 2 > 1$ when $k \geq 2$. If $y = [c_{j+1}, \ldots, c_k]^- > 1$ and $j \geq 2$, then $[c_j, \ldots, c_k]^- = c_j - 1/y > c_j - 1 \geq 1$.

Uniqueness. Take any representation $x = [c_1, \ldots, c_k]^-$ with these digit conditions. If $k = 1$ then $x = c_1 \in \mathbb{Z}$. If $k \geq 2$ then $x = c_1 - 1/y$ with $y = [c_2, \ldots, c_k]^- > 1$, so $x \in (c_1 - 1, c_1)$ is not an integer, $c_1 = \lceil x \rceil$, and $y = 1/(c_1 - x)$. So an integer has only the representation with $k = 1$, and for a non-integer the first digit and the value y of the tail are determined. The tail is a representation of y of the same kind, so by induction on k it is unique too. □

Theorem 4.2 (distance formula, negative-continued-fraction form)**.** *Let $x > 0$ have the expansion of Lemma 4.1. Then*

$$d(x) = c_1 + c_2 + \cdots + c_k + k - 1.$$

Together with $d(0) = 0$ and $d(x) = d(-1/x) + 1$ for $x < 0$, which is (D2), this gives d on all of $\mathbb{Q}$.

Proof. By Theorem 3.6 we can compute D instead. We use induction on k. If $k = 1$ then $x = c_1$ is a positive integer, and its parent sequence is c_1 subtractions. Now let $k \geq 2$. Since $c_1 = \lceil x \rceil$, the sequence first subtracts 1 a total of c_1 times and gets to

$$x - c_1 = -\frac{1}{[c_2, \ldots, c_k]^-} \in (-1, 0),$$

which is in $(-1, 0)$ because the tail is bigger than 1. One flip then gives $[c_2, \ldots, c_k]^-$, whose digits satisfy the conditions of Lemma 4.1. By induction the rest of the sequence takes $\left(\sum_{i\geq 2} c_i\right)+(k-1)-1$ steps, so the total is $c_1 + 1 + \sum_{i\geq 2} c_i + k - 2 = \sum_i c_i + k - 1$. □

Corollary 4.3 (the ranks explicitly, answering Q3)**.** *For $n \geq 1$,*

$$X_n \cap \mathbb{Q}_{>0} = \left\{ [c_1, \ldots, c_k]^- \;\middle|\; k \geq 1,\ c_1 \geq 1,\ c_2, \ldots, c_k \geq 2,\ \textstyle\sum_i c_i = n + 1 - k \right\},$$
$$X_n \cap \mathbb{Q}_{<0} = \left\{ -1/y \;\middle|\; y \in X_{n-1} \cap \mathbb{Q}_{>0} \right\},$$

and in both lines each element is listed once, since different digit strings give different rationals and $y \mapsto -1/y$ is injective. In particular rank n is finite, and its elements correspond to integer compositions.

In words: put $s_1 = c_1$ and $s_i = c_i + 1$ for $i \geq 2$. Then the positive rationals of rank n match the ordered sums $n = s_1 + \cdots + s_k$ with $s_1 \geq 1$ and $s_2, \ldots, s_k \geq 3$, and the negative ones are $-1/y$ for the positive y one rank lower. There are finitely many such sums, so X_n is finite.

Proof. The first line follows from Lemma 4.1, which says each positive rational has exactly one digit string, and Theorem 4.2, which says its rank is $\sum c_i + k - 1$. For the second line, if $x < 0$ then $-1/x > 0$ and (D2) gives $d(-1/x) = d(x) - 1$. Conversely, if $y > 0$ then $-1/y < 0$ and (D3) gives $d(-1/y) = d(y) + 1$. So $x \mapsto -1/x$ is a bijection from $X_n \cap \mathbb{Q}_{<0}$ to $X_{n-1} \cap \mathbb{Q}_{>0}$. Finally $X_n \subseteq \mathbb{Q}_{>0} \cup \mathbb{Q}_{<0}$ for $n \geq 1$, and each $c_i \geq 1$ forces $k \leq n$ and $c_i \leq n$, so both lines are finite. □

Example 4.4. Take $n = 5$. The digit strings with $\sum_i c_i + k - 1 = 5$ are (5), $(1, 3)$ and $(2, 2)$, with values 5, $[1, 3]^- = \frac{2}{3}$ and $[2, 2]^- = \frac{3}{2}$. A string of length 3 would need $c_1 + c_2 + c_3 = 3$, which is impossible with $c_2, c_3 \geq 2$. The positive rationals of rank 4 are 4 and $\frac{1}{2}$, from the strings (4) and $(1, 2)$, and they give the negatives $-\frac{1}{4}$ and -2. So $X_5 = \{5, \frac{2}{3}, \frac{3}{2}, -\frac{1}{4}, -2\}$, as in Table 1.

Example 4.5. For $x = \frac{2}{7}$ the algorithm in the proof of Lemma 4.1 passes through $\frac{2}{7}, \frac{7}{5}, \frac{5}{3}, 3$ and gives $\frac{2}{7} = [1, 2, 2, 3]^-$. So $d(2/7) = (1 + 2 + 2 + 3) + 4 - 1 = 11$ and $d(-7/2) = 12$, as in Example 3.10.

4.1 The ordinary continued fraction form

The ordinary (regular) continued fraction is more familiar. Recall the notation

$$[b_0; b_1, \ldots, b_m] = b_0 + \cfrac{1}{b_1 + \cfrac{1}{\ddots + \cfrac{1}{b_m}}}, \qquad b_0 \geq 0,\ b_i \geq 1\ (1 \leq i \leq m),\ b_i \in \mathbb{Z}.$$

We call m the *length*, and a finite sequence of integers ≥ 1 (possibly empty) a *tail*. It is well known that a positive rational has exactly 2 such expansions [11, Ch. X], of lengths differing by 1. We prove the version we need to keep the paper self-contained.

Here is where the weights 1 and 3 in Theorem B(iii) come from. One can check that if $x = [b_0; b_1, \ldots, b_m]$ with m odd, then the negative continued fraction of x is

$$[\, b_0 + 1,\ 2^{(b_1-1)},\ b_2 + 2,\ 2^{(b_3-1)},\ \ldots,\ b_{m-1} + 2,\ 2^{(b_m-1)}\,]^-,$$

where $2^{(j)}$ means j digits equal to 2; for example $\frac{2}{7} = [0; 3, 1, 1]$ gives $[1, 2, 2, 3]^-$. By Theorem 4.2 a digit 2 costs 3 moves (2 subtractions and 1 flip), so each unit of an odd-position digit costs 3, and each unit of an even-position digit costs 1. We do not use this conversion: the proof below checks directly that the formula satisfies the same recursion as D.

Lemma 4.6 (odd-length normalization)**.** *(a) For an expansion as above with $m \geq 1$, the value $v = [b_1; b_2, \ldots, b_m]$ of the tail satisfies $v \geq 1$, with $v = 1$ exactly when $m = 1$ and $b_1 = 1$.*

(b) Every rational $x > 0$ has exactly one expansion $x = [b_0; b_1, \ldots, b_m]$ of odd *length m.*

(c) In the odd-length expansion, $b_0 \geq 1$ if and only if $x > 1$.

Proof. (a) If $m = 1$ then $v = b_1 \geq 1$, with equality if and only if $b_1 = 1$. If $m \geq 2$ then $v = b_1 + 1/[b_2; \ldots, b_m] > b_1 \geq 1$, since $[b_2; \ldots, b_m] \geq b_2 > 0$.

(b) We show, by strong induction on the denominator, that every rational $x > 0$ has exactly one expansion of each parity of the length. Let $x > 0$ be arbitrary.

Case 1: $x \in \mathbb{Z}$. The only expansion of length 0 is $[x]$. An expansion of length $m \geq 1$ has $x = b_0 + 1/v$ with $v \geq 1$ by (a). Then $1/v = x - b_0$ is an integer in $(0, 1]$, so $v = 1$, which by (a) means $m = 1$ and $b_1 = 1$, and then $b_0 = x - 1 \geq 0$. So x has exactly the expansions $[x]$ and $[x - 1; 1]$, one of each parity. This case includes all x with denominator 1.

Case 2: $x \notin \mathbb{Z}$. There is no expansion of length 0. An expansion of length $m \geq 1$ has $x = b_0 + 1/v$ with $v = [b_1; \ldots, b_m] \geq 1$, and $v \neq 1$ because x is not an integer. So $0 < 1/v < 1$, which forces $b_0 = \lfloor x \rfloor$ and $v = 1/(x - \lfloor x \rfloor)$. Conversely, every expansion of v has first digit at least 1, because an expansion with first digit 0 has value at most 1 by (a), and $v > 1$. So the expansions of x of length m are exactly the expansions of v of length $m - 1$ with $\lfloor x \rfloor$ put in front. If $x = p/q$, then $v = q/(p - \lfloor x \rfloor q)$ is in lowest terms with denominator $p - \lfloor x \rfloor q < q$. By induction v has exactly one expansion of each parity, and hence so does x.

(c) Since m is odd, $m \geq 1$, so $x = b_0 + 1/v$ with $v \geq 1$. If $b_0 \geq 1$ then $x > b_0 \geq 1$, and if $b_0 = 0$ then $x = 1/v \leq 1$. Hence $b_0 \geq 1$ exactly when $x > 1$. Note that $x = 1$ itself has the odd-length form $[0; 1]$. □

Lemma 4.7 (complement identity)**.** *Let $b \geq 1$, let* t *be a tail, possibly empty, and put $u = [b; \mathrm{t}] \geq 1$. Then*

$$[0; 1, b, \mathrm{t}] = \frac{u}{1+u}, \qquad [0; b+1, \mathrm{t}] = \frac{1}{1+u},$$

and in particular $[0; 1, b, \mathrm{t}] + [0; b+1, \mathrm{t}] = 1$.

Proof. We have $[1; b, \mathrm{t}] = 1 + 1/u = (1+u)/u$, so $[0; 1, b, \mathrm{t}] = u/(1+u)$. Also $[b+1; \mathrm{t}] = 1 + [b; \mathrm{t}] = 1 + u$, so $[0; b+1, \mathrm{t}] = 1/(1+u)$. □

Theorem 4.8 (distance formula, ordinary-continued-fraction form)**.** *Let $x > 0$ and let $x = [b_0; b_1, \ldots, b_m]$ be its unique odd-length expansion (Lemma 4.6). Then*

$$d(x) \;=\; \sum_{i \text{ even}} b_i \;+\; 3 \sum_{i \text{ odd}} b_i \;-\; 2.$$

In words: in the odd-length expansion, each unit of a digit in an even position costs 1 move and each unit of a digit in an odd position costs 3 moves, and then we subtract 2. For example $\frac{2}{7} = [0;3,2] = [0;3,1,1]$ has rank $(0+1)+3(3+1)-2 = 11$, as in Example 4.5, and $\frac{3}{5} = [0;1,1,2]$ has rank $(0+1)+3(1+2)-2 = 8$.

Proof. Step 1: reduction. Let $E(x)$ be the right-hand side for $x > 0$, and extend E to all of $\mathbb{Q}$ by $E(0) = 0$ and $E(x) = E(-1/x) + 1$ for $x < 0$. By the uniqueness principle (Lemma 3.4) and Theorem 3.6, it is enough to show $E(x) = E(P(x)) + 1$ for every $x \neq 0$. For $x < 0$ this is the definition, so there are 3 cases left, all with $x > 0$.

Step 2: $x > 1$. By Lemma 4.6(c), $b_0 \geq 1$, and $x - 1 = [b_0 - 1; b_1, \ldots, b_m]$ is again an odd-length expansion, where the first digit may now be 0. By uniqueness it is *the* odd-length expansion of $x - 1 > 0$. Only the digit b_0, in an even position, changed, so $E(x) = E(x-1) + 1$.

Step 3: $x = 1$. Here $E(1) = E([0;1]) = 0 + 3 \cdot 1 - 2 = 1 = E(0) + 1$.

Step 4: $0 < x < 1$. Here $b_0 = 0$ and $P(x) = x - 1 \in (-1, 0)$, so $E(x-1) = E\big(\frac{1}{1-x}\big) + 1$, and we have to show $E(x) = E\big(\frac{1}{1-x}\big) + 2$. Note that $m \geq 1$, and if $b_1 = 1$ then $m \geq 3$, since $[0;1] = 1$ is not in this case. We use Lemma 4.7.

- Suppose $b_1 = 1$, so $x = [0;1,b_2,\mathrm{t}]$ with tail $\mathrm{t} = b_3, \ldots, b_m$. Then $1 - x = [0; b_2 + 1, \mathrm{t}]$, and hence
$$\tfrac{1}{1-x} = [\, b_2 + 1;\, b_3, \ldots, b_m \,],$$
which is an expansion of odd length $m - 2$. Compared with the digits of x, position 0 went up by 1, which has weight 1, and the digit $b_1 = 1$ in an odd position is gone, which had weight 3. All the other digits moved by 2 places, so they kept the parity of their position. Hence $E\big(\frac{1}{1-x}\big) = E(x) + 1 - 3 = E(x) - 2$.
- Suppose $b_1 \geq 2$, so $x = [0; (b_1 - 1) + 1, \mathrm{t}]$ with tail $\mathrm{t} = b_2, \ldots, b_m$. Then $1 - x = [0; 1, b_1 - 1, \mathrm{t}]$, and hence
$$\tfrac{1}{1-x} = [\, 1;\, b_1 - 1,\, b_2, \ldots, b_m \,],$$
which has the same odd length m. Position 0 went up by 1 and position 1 went down by 1, which has weight 3, and nothing else moved. Hence again $E\big(\frac{1}{1-x}\big) = E(x) + 1 - 3 = E(x) - 2$.

In both cases the new expansion is the odd-length expansion of $\frac{1}{1-x}$, by uniqueness, so the digit count above computes $E\big(\frac{1}{1-x}\big)$. □

5 Counting the ranks (Q1)

Here is why $a(n-3)$ appears. Corollary 3.8(c) says where the children of the 4 classes $(1,\infty)$, $(0,1]$, $(-1,0)$, $(-\infty,-1]$ go (Figure 2). A positive z produces new positive rationals in 2 ways: $z+1$ a rank later, and the end of the detour $z \to z+1 \to -1/(z+1) \to z/(z+1)$ (moves f, g, f) 3 ranks later. After a g-move this detour is the only way back to the positives. So the positive counts satisfy $r(n) = r(n-1) + r(n-3)$, and the negative rationals copy the positive ones a rank later. Adding these gives Kagey's recurrence.

Theorem 5.1 (main count)**.** *Recall $r(n) = |X_n \cap \mathbb{Q}_{>0}|$. Then $a(0) = 1$, $a(n) = r(n) + r(n-1)$ for $n \geq 1$, and, as formal power series,*

$$R(t) := \sum_{n \geq 1} r(n)\, t^n = \frac{t}{1 - t - t^3}, \qquad A(t) := \sum_{n \geq 0} a(n)\, t^n = \frac{1 + t^2 - t^3}{1 - t - t^3}.$$

Consequently

(a) $a(n) = a(n-1) + a(n-3)$ *for every* $n \geq 4$,

(b) the recurrence fails at $n = 3$, *where* $a(3) - a(2) - a(0) = -1$, *and*

(c) $r(n) = \text{A000930}(n-1)$ (Narayana's cows) *and* $a(n) = \text{A097333}(n-1)$ *for* $n \geq 1$.

In words: this answers Q1, and $a(n)$ is the sequence that Kimberling's entry names.

Proof. We set $r(m) = 0$ for $m \leq 0$; this agrees with the definition for $m = 0$, since $X_0 = \{0\}$.

Step 1: negatives. For $n \geq 1$, X_n contains no 0, and $x \mapsto -1/x$ is a bijection from $X_n \cap \mathbb{Q}_{<0}$ to $X_{n-1} \cap \mathbb{Q}_{>0}$ (proof of Corollary 4.3). So $a(n) = r(n) + r(n-1)$.

Step 2: positives. We show $r(n) = r(n-1) + r(n-3)$ for all $n \geq 2$. Let $n \geq 2$. Since $d(1) = 1$ $(1 = f(0))$, every $x \in X_n \cap \mathbb{Q}_{>0}$ lies in $(1, \infty)$ or in $(0, 1)$.

- The map $x \mapsto x - 1$ is a bijection from $X_n \cap (1, \infty)$ to $X_{n-1} \cap \mathbb{Q}_{>0}$. Indeed, if $x > 1$ then $x - 1 > 0$ and $d(x-1) = d(x) - 1$ by (D1). Conversely, if $y > 0$ then $y + 1 > 1$ and $d(y+1) = d(y) + 1$ by (D1).
- The map $x \mapsto x/(1-x)$ is a bijection from $X_n \cap (0, 1)$ to $X_{n-3} \cap \mathbb{Q}_{>0}$, with inverse $z \mapsto z/(1+z)$. Indeed, let $x \in (0, 1)$. Then $d(x) = d(x-1)+1$ by (D1). Next $x-1 \in (-1, 0)$, so $d(x-1) = d\big(\frac{1}{1-x}\big)+1$ by (D2). Finally $\frac{1}{1-x} > 1$ and $\frac{1}{1-x} - 1 = \frac{x}{1-x} > 0$, so $d\big(\frac{1}{1-x}\big) = d\big(\frac{x}{1-x}\big)+1$ by (D1). Together, $d\big(\frac{x}{1-x}\big) = d(x) - 3$. Conversely, every $z > 0$ is $\frac{x}{1-x}$ for $x = \frac{z}{1+z} \in (0, 1)$, and then $d(x) = d(z) + 3$.

Adding the 2 counts gives $r(n) = r(n-1) + r(n-3)$ (for $n = 2, 3$ the second set is empty).

Step 3: the recurrence. Let $n \geq 4$. Using Step 1 at n, $n-1$ and $n-3$ (all ≥ 1) and Step 2 at n and $n-1$ (both ≥ 2),

$$a(n) = r(n) + r(n-1) = \big[r(n-1) + r(n-3)\big] + \big[r(n-2) + r(n-4)\big] = a(n-1) + a(n-3).$$

This is (a). For (b), $a(3) = 2$ while $a(2) + a(0) = 3$ (Table 1).

Step 4: generating functions. The coefficient of t^n in $(1 - t - t^3)R(t)$ is $r(n) - r(n-1) - r(n-3)$. It is 0 for $n \geq 2$ by Step 2, it is $r(1) = 1$ for $n = 1$, and it is 0 for $n = 0$. So $(1 - t - t^3)R(t) = t$, which is the formula for R. By Step 1,

$$A(t) = 1 + R(t) + tR(t) = 1 + \frac{(1+t)t}{1 - t - t^3} = \frac{(1 - t - t^3) + t + t^2}{1 - t - t^3} = \frac{1 + t^2 - t^3}{1 - t - t^3}.$$

For (c), A000930 has generating function $1/(1 - t - t^3)$ and A097333 has generating function $(1 + t)/(1 - t - t^3)$ [8, 7], so $R(t) = t \cdot \frac{1}{1-t-t^3}$ and $A(t) - 1 = t \cdot \frac{1+t}{1-t-t^3}$ give the stated shifts. □

Another way to see $R(t)$**.** Corollary 4.3 connects Q3 with Q1. By it, $r(n)$ counts the compositions of n whose first part is ≥ 1 and whose other parts are ≥ 3. The first part contributes $t/(1-t)$ and each other part $t^3/(1-t)$, so $R(t) = \sum_{k \geq 1} \frac{t}{1-t}\big(\frac{t^3}{1-t}\big)^{k-1} = \frac{t}{1-t-t^3}$, a geometric series that makes sense formally because $t^3/(1-t)$ has no constant term.

Proposition 5.2 (asymptotics)**.** *Let* $\psi = 1.4655712318\ldots$ *be the real root of* $\psi^3 = \psi^2 + 1$. *Then*

$$a(n) = C\,\psi^n + O\big(\psi^{-n/2}\big), \qquad C = \frac{\psi^2 + \psi}{\psi^2 + 3} = 0.7019310679\ldots,$$

with an absolute implied constant.

Proof. The roots. The polynomial $h(x) = x^3 - x^2 - 1$ has derivative $3x^2 - 2x$, which vanishes at 0 and $\frac{2}{3}$, where h is -1 and $-\frac{31}{27}$. So h has exactly one real root ψ, which lies in $(1, 2)$ since $h(1) < 0 < h(2)$, and a pair of complex conjugate roots $\lambda, \bar{\lambda}$. By Vieta $\psi\,|\lambda|^2 = 1$, so $|\lambda| = \psi^{-1/2} < 1$. Since $Q(t) := 1 - t - t^3 = t^3 h(1/t)$, the roots of Q are the 3 distinct, hence simple, numbers $t_0 = 1/\psi$, $1/\lambda$ and $1/\bar{\lambda}$.

Partial fractions. Since $1 + t^2 - t^3 = Q(t) + t + t^2$, we have $A(t) = 1 + (t + t^2)/Q(t)$. A simple pole at t_1 with residue ρ contributes $\rho/(t - t_1) = -(\rho/t_1) \sum_n t_1^{-n} t^n$, so for every $n \geq 1$

$$a(n) = C\, t_0^{-n} + c\, \lambda^n + \bar{c}\, \bar{\lambda}^{\,n}, \qquad C = \frac{t_0 + t_0^2}{t_0\,\big(-Q'(t_0)\big)},$$

for some constant c. Since $t_0^{-n} = \psi^n$ and $|\lambda^n| = \psi^{-n/2}$, this proves the asymptotic formula, with implied constant $\max\big(2|c|, |a(0) - C|\big)$. Finally $-Q'(t) = 1 + 3t^2$, so

$$C = \frac{t_0 + t_0^2}{t_0\,(1 + 3t_0^2)} = \frac{1 + t_0}{1 + 3t_0^2} = \frac{\psi^2 + \psi}{\psi^2 + 3},$$

multiplying numerator and denominator by ψ^2 and using $t_0 = 1/\psi$. □

Proposition 5.3 (the four-class transfer structure)**.** *For $n \geq 1$ put*

$$\alpha_n = |X_n \cap (1, \infty)|, \quad \beta_n = |X_n \cap (0, 1]|, \quad \gamma_n = |X_n \cap (-1, 0)|, \quad \delta_n = |X_n \cap (-\infty, -1]|.$$

Then for all $n \geq 1$,

$$\alpha_{n+1} = \alpha_n + \beta_n, \qquad \beta_{n+1} = \gamma_n, \qquad \gamma_{n+1} = \alpha_n, \qquad \delta_{n+1} = \beta_n.$$

The transfer matrix has characteristic polynomial $t\,(t^3 - t^2 - 1)$.

Proof. By Corollary 3.8, X_{n+1} is the set of children of the elements of X_n, and each element of X_{n+1} is the child of exactly one of them. The f-child $x + 1$ lies in $(1, \infty)$ if $x > 0$ and in $(0, 1)$ if $x \in (-1, 0)$. The g-child $-1/x$ of a positive x lies in $(-1, 0)$ if $x > 1$ and in $(-\infty, -1]$ if $x \leq 1$. Vertices ≤ -1 have no children. Counting the children that land in each class gives the 4 rules. The matrix of the linear map $(\alpha, \beta, \gamma, \delta) \mapsto (\alpha + \beta,\ \gamma,\ \alpha,\ \beta)$ is

$$M = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \det(tI - M) = t\,(t^3 - t^2 - 1),$$

by expanding along the last column, which is zero in M because the δ-class has no children. □

6 Geodesic words, and completion of the transfer-matrix argument

This section is not needed for Theorems A–C. It compares the parent-map words with [3].

Proposition 6.1 (canonical normal forms)**.** *For $x \in \mathbb{Q}$ let $w(x)$ be the word we read along the reversed parent sequence of x, as in Example 3.10, with $w(0)$ empty. Then*

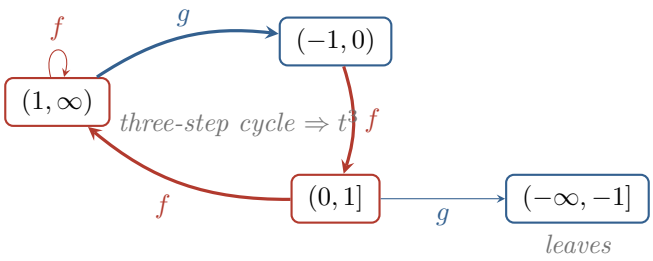


Figure 2: Where the children of each of the 4 classes go (Proposition 5.3). After a g-step, the only way back to the positives is the three-step cycle $(1,\infty) \to (-1,0) \to (0,1] \to (1,\infty)$. This cycle gives the t^3 in the recurrence, and it is the counting version of $(gf)^3 = \mathrm{id}$ (Remark 3.9).

(a) $w(x)$ *is admissible at* 0*, ends at* x*, and has length* $|w(x)| = d(x)$*,*

(b) $x \mapsto w(x)$ *is a bijection from* $\mathbb{Q}$ *onto the language*

$$\mathcal{L} \;=\; \{\varepsilon\} \;\cup\; \Big\{ f^{m_1} g\, f^{m_2} g \cdots f^{m_j} g^{\epsilon} \;\Big|\; j \ge 1,\ \epsilon \in \{0,1\},\ m_1, \ldots, m_{j-1} \ge 2,\ m_j \ge 1 \Big\},$$

where ε *is the empty word,*

(c) under this bijection $\epsilon = 1$ *exactly when* $x < 0$*, and for* $x > 0$ *the run lengths are the digits of the negative continued fraction in reverse order,* $(m_1, \ldots, m_j) = (c_k, \ldots, c_1)$*.*

Proof. Part (a) is Step 1 of the proof of Theorem 3.6, together with $d = D$. For (b) and (c), applying $w(x)$ to 0 gives back x, so the map is injective. Let $x > 0$ have digits $(c_1, \ldots, c_k)$. By the proof of Theorem 4.2, its parent sequence is c_1 subtractions, a flip, c_2 subtractions, and so on, ending with c_k subtractions. Reversing this, the word is $f^{c_k} g\, f^{c_{k-1}} g \cdots g\, f^{c_1}$, so the runs are $(m_1, \ldots, m_j) = (c_k, \ldots, c_1)$ with $j = k$ and $\epsilon = 0$. The conditions $c_1 \ge 1$ and $c_i \ge 2$ for $i \ge 2$ become $m_j \ge 1$ and $m_i \ge 2$ for $i \le j-1$. For $x < 0$ we have $w(x) = w(-1/x)\, g$, which has the same runs as $w(-1/x)$ and $\epsilon = 1$. Conversely, given $(j, \epsilon, m_1, \ldots, m_j)$ allowed in $\mathcal{L}$, let $y = [m_j, \ldots, m_1]^- > 0$. Then $w(y)$ has runs $(m_1, \ldots, m_j)$, so the word is $w(y)$ if $\epsilon = 0$ and $w(-1/y)$ if $\epsilon = 1$. (By Lemma 4.1, $(m_j, \ldots, m_1)$ is the digit string of y, so this follows from the first part of the proof.) □

Proposition 6.2 (pattern-avoidance description)**.** $\mathcal{L}$ *equals the set of words over* $\{f, g\}$ *containing no factor* gg*, containing no factor* $fgfgf$*, not beginning with* g*, and not beginning with* fgf*.*

Proof. The empty word is in both sets. A nonempty word with no gg that does not start with g can be written in exactly one way as

$$f^{m_1} g\, f^{m_2} g \cdots f^{m_j} g^{\epsilon}, \qquad j \ge 1,\ m_i \ge 1,\ \ \epsilon \in \{0,1\},$$

since every g is preceded by an f and the word may end in g. In such a word, an occurrence of $fgfgf$ has its middle f forming a whole run of length 1, with a g on each side and an f after the right-hand g. A run m_i has a g on each side when $2 \le i \le j-1$, or when $i = j \ge 2$ and $\epsilon = 1$, but in the second case nothing comes after the last g. So having no $fgfgf$ is the same as $m_i \ge 2$ for $2 \le i \le j-1$. In the same way, the word starts with fgf exactly when $m_1 = 1$ and $j \ge 2$. Putting these together, the 4 conditions say $m_i \ge 2$ for $1 \le i \le j-1$ and $m_j \ge 1$, which is the definition of $\mathcal{L}$. □

Remark 6.3 (the Math.SE argument). The answer [3] showed that shortest words avoid the 4 patterns of Proposition 6.2 (using $g^2 = \mathrm{id}$, $fgfgf = g$ and the start at 0), and counted the words of $\mathcal{L}$ with a 14-state automaton. Proposition 6.1 adds what was missing: *every* word of $\mathcal{L}$ is a shortest word, and $\mathcal{L}$ matches $\mathbb{Q}$ one to one. So that count is a second proof of Theorem 5.1(a). The script `verify.py` also checks that the tree words, $\mathcal{L}$ and the pattern-avoiding words are the same up to length 14 (Section 7).

7 Computational verification

The script `verify.py` (plain Python, exact integer arithmetic) runs Kimberling's construction with each rational stored as a reduced pair (p, q), which $f(p/q) = (p+q)/q$ and $g(p/q) = -q/p$ keep reduced. Up to rank 41 (14,144,885 vertices) it checks the following, and none of the checks failed.

1. The rank sizes $a(0), \ldots, a(41)$ in Table 2 satisfy $a(n) = a(n-1) + a(n-3)$ for $4 \le n \le 41$, fail it at $n = 3$, and agree with A097333 shifted by one.
2. At every vertex, the map that produced it is f exactly when the value is positive (Q2).
3. Corollaries 3.7 and 3.8: $d(P(x)) = d(x) - 1$, children have rank $d(x) + 1$ (below rank 41), $d(x+1) = d(x) - 3$ for $x < -1$, and no value appears twice in a generation.
4. The rank formulas of Theorems 4.2 and 4.8 hold at all 453,456 nonzero vertices of rank at most 32, with both continued fractions computed directly from p/q, not from the search.
5. The rules of Proposition 5.3 hold for $1 \le n \le 40$.
6. The rows, in order, agree with the first 73 terms of A226248 and A226247 as stored in the script (ranks 0 to 9 and the start of rank 10).
7. Propositions 6.1 and 6.2 up to length 14: the words read off the tree, the words of $\mathcal{L}$ and the words avoiding the 4 patterns are the same 465 words, and each of them, applied to 0, gives a different rational whose rank is its length.

n	$a(n)$	n	$a(n)$	n	$a(n)$	n	$a(n)$	n	$a(n)$	n	$a(n)$
0	1	7	10	14	148	21	2,150	28	31,224	35	453,458
1	1	8	15	15	217	22	3,151	29	45,761	36	664,575
2	2	9	22	16	318	23	4,618	30	67,066	37	973,982
3	2	10	32	17	466	24	6,768	31	98,290	38	1,427,440
4	3	11	47	18	683	25	9,919	32	144,051	39	2,092,015
5	5	12	69	19	1,001	26	14,537	33	211,117	40	3,065,997
6	7	13	101	20	1,467	27	21,305	34	309,407	41	4,493,437

Table 2: Verified rank sizes $a(n) = |X_n|$ for $0 \le n \le 41$.

8 Further remarks

Remark 8.1 (negative integers). For $j \ge 1$, induction on j shows that $[2, \ldots, 2]^-$ with j twos is $(j+1)/j$. So $\frac{1}{n} = [1, 2^{(n-1)}]^-$, one 1 followed by $n-1$ twos, for every $n \ge 1$. Hence $d(1/n) = \big(1 + 2(n-1)\big) + n - 1 = 3n - 2$ and $d(-n) = 3n - 1$. Thus $-1, -2, -3, \ldots$ appear at ranks $2, 5, 8, \ldots$, 3 apart, which again comes from the three-step cycle.

Remark 8.2 (sharpness at $n = 3$). The one failure, $a(3) = a(2) + a(0) - 1$, comes from the root. Step 3 of the proof of Theorem 5.1 writes $a(n-3)$ as $r(n-3) + r(n-4)$, which needs $n - 3 \geq 1$; at $n = 3$ the term $a(0) = 1$ counts the root, which is neither positive nor of the form $-1/y$. Put another way, the detour f, g, f takes a positive z to the new vertex $z/(1+z)$, but takes the root back to 0. In Section 6 this is the rule that a word may not begin with fgf.

Remark 8.3 (positions of the nonpositive entries). If we list Kimberling's rows one after another, the nonpositive values are at positions $n + \mathrm{A005374}(n)$, where A005374 is Hofstadter's H-sequence, as Gómez Calderón noted in A226249 [6]. We do not prove this, but it fits with Proposition 5.3, whose counts grow by the rule $t^3 = t^2 + 1$ that is behind A005374.

Remark 8.4 (variants). Kimberling's entry also has a second ordering S' of the rationals, built from the same 2 maps with different conventions (A226130), with a conjectured count in A226136 (see also [2]). The parent-map method should work there too: guess P from the small ranks, then prove the versions of (D1)–(D4).

Data and code availability

The script `verify.py` (about 1–2 minutes, 2 GB of memory) and the solution notes are in the Problem 137 folder of `https://github.com/apemm/Kagey-Problems`. In the same repository, the Lean 4 file `lean/KageyProblems/Problem137.lean` formalizes Lemma 3.2, Theorem 3.6, Corollaries 3.7–3.8, Theorem 4.2, Corollary 4.3 (without the uniqueness in Lemma 4.1), Proposition 5.3, Theorem 5.1(a)–(c) in recurrence form, and $d(-n) = 3n - 1$.

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